

Course: BCA/BCA-DS

Semester/Year: Vth/IIIrd

Session: 2025-26

Subject Name & Code: Soft Computing [BCAC0028]

Section: DS-A, C, E

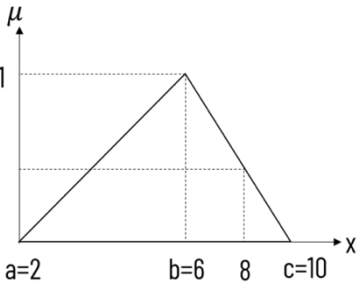
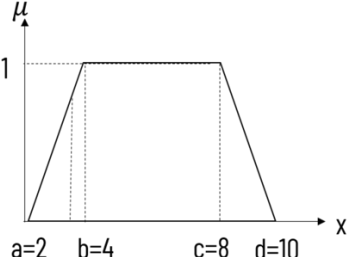
Maximum Marks: 10

Submission Date: 08/12/2025

Important Instructions:

- All answers must be **handwritten** on **A4-sized paper** using only **blue or black ink pens** for writing.
- Clearly write the **question number and the question** before providing the answer.
- Include **neat, accurate, and clearly labeled diagrams** wherever required, ensuring they are drawn with appropriate tools for precision.
- For numerical problems, **show each step clearly** with proper calculations, units, and explanations to ensure full transparency.

Q.No.	Question																					
1.	List and Explain the basic fuzzy operations: Union, Intersection, and Complement with proper examples.																					
2.	What is a fuzzy membership function? Explain its purpose with examples. Draw and describe any two common types of membership functions used in fuzzy logic (such as triangular or trapezoidal).																					
3.	Explain how a fuzzy relation is represented using a membership matrix.																					
4.	Explain the features of fuzzy membership functions with proper diagram.																					
5.	Using your own intuition and definitions of the universe of discourse plot fuzzy membership function for Weight and Height of the people.																					
6.	We want to compare two software performance controllers for their response levels under different CPU load percentages. The response level indicates how efficiently each controller scales or reacts to the system load. The following values of CPU load and corresponding controller response levels were recorded using a performance monitoring tool: <table><tr><th>CPU Load(%)</th><th>Response Level 1</th><th>Response Level 2</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>20</td><td>0.5</td><td>0.45</td></tr><tr><td>40</td><td>0.35</td><td>0.55</td></tr><tr><td>60</td><td>0.75</td><td>0.65</td></tr><tr><td>80</td><td>0.95</td><td>0.9</td></tr><tr><td>100</td><td>1</td><td>1</td></tr></table> Given the universe of discourse is $X=\{0,20,40,60,80,100\}$ and the membership functions $C_1 = \left\{ \frac{0}{0} + \frac{0.5}{20} + \frac{0.35}{40} + \frac{0.75}{60} + \frac{0.95}{80} + \frac{1}{100} \right\}$ $C_2 = \left\{ \frac{0}{0} + \frac{0.45}{20} + \frac{0.55}{40} + \frac{0.65}{60} + \frac{0.9}{80} + \frac{1}{100} \right\}$ Compute the following $\mu_{C_1 \cup C_2}(x)$ $\mu_{C_1 \cap C_2}(x)$ $\mu_{\overline{C_1}}(x)$ $\mu_{\overline{C_2}}(x)$ $\mu_{C_1 \cup \overline{C_1}}(x)$ $\mu_{C_2 \cap \overline{C_2}}(x)$ $\mu_{C_1 C_2}(x)$ $\mu_{C_2 C_1}(x)$	CPU Load(%)	Response Level 1	Response Level 2	0	0	0	20	0.5	0.45	40	0.35	0.55	60	0.75	0.65	80	0.95	0.9	100	1	1
CPU Load(%)	Response Level 1	Response Level 2																				
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7.	Consider two given fuzzy sets A and B, perform union, intersection, complement, bounded-sum, bounded-difference, algebraic-sum and algebraic-product: $A = \left\{ \frac{1}{2} + \frac{0.3}{4} + \frac{0.5}{6} + \frac{0.2}{8} \right\}$ $B = \left\{ \frac{0.5}{2} + \frac{0.4}{4} + \frac{0.1}{6} + \frac{1}{8} \right\}$																					

8.	$X = \{x_1, x_2\}$, $Y = \{y_1, y_2\}$, and $Z = \{z_1, z_2, z_3\}$. Consider the following fuzzy relations: Relation $\underline{R}$$\bar{R} = \begin{matrix} & y_1 & y_2 \\ x_1 & [0.7 & 0.6] \\ x_2 & [0.8 & 0.3] \end{matrix}$ Relation $\underline{S}$$\bar{S} = \begin{matrix} & z_1 & z_2 & z_3 \\ y_1 & [0.8 & 0.5 & 0.4] \\ y_2 & [0.1 & 0.6 & 0.7] \end{matrix}$ 1: Perform the MAXMIN and MAX-PRODUCT composition operation on the fuzzy relation.																			
9.	Calculate fuzzy membership value for the following : 1: Determine μ, corresponding to $x = 8.0$  2: Determine μ, corresponding to $x = 3.5$ 																			
10.	What is defuzzification? Differentiate b/w fuzzification and defuzzification.																			
11.	1. Consider two fuzzy sets $\underline{A}$ and $\underline{B}$, both defined on X, given as follows: <table><tr><th>$\mu(x_i; X)$</th><th>x_1</th><th>x_2</th><th>x_3</th><th>x_4</th><th>x_5</th></tr><tr><td>$\underline{A}$</td><td>0.2</td><td>0.3</td><td>0.4</td><td>0.7</td><td>0.1</td></tr><tr><td>$\underline{B}$</td><td>0.4</td><td>0.5</td><td>0.6</td><td>0.8</td><td>0.9</td></tr></table> Express the following λ-cut sets using Zadeh's notation: (a) $(\underline{A})_{0.7}$; (b) $(\underline{B})_{0.2}$; (c) $(\underline{A} \cup \underline{B})_{0.6}$; (d) $(\underline{A} \cap \underline{B})_{0.5}$; (e) $(\underline{A} \cup \underline{A})_{0.7}$; (f) $(\underline{B} \cap \underline{B})_{0.3}$; (g) $(\underline{A} \cap \underline{B})_{0.6}$; (h) $(\underline{A} \cup \underline{B})_{0.8}$		$\mu(x_i; X)$	x_1	x_2	x_3	x_4	x_5	$\underline{A}$	0.2	0.3	0.4	0.7	0.1	$\underline{B}$	0.4	0.5	0.6	0.8	0.9
$\mu(x_i; X)$	x_1	x_2	x_3	x_4	x_5															
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12.	Discuss the basic terminologies(Individuals, Genes, Fitness and Populations) of Genetic Algorithm(GA) and mention key difference between Genetic and Traditional algorithms.																			
13.	Describe the encoding operator in Genetic Algorithm(GA). Explain binary, octal, hexadecimal, permutation, value and tree encoding schemes with proper examples.																			
14.	List and briefly explain the main steps involved in a Genetic Algorithm(GA) and describe its workflow.																			
15.	Explain the crossover process in Genetic Algorithm(GA). Describe at least three common crossover methods.																			